



Seasonal Adjustment

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■ Agenda

■ Motivation

■ Chinese holidays

■ Construction of regressors

■ Reminder: RegARIMA

■ Results

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I Motivation

- Frequently **no official** seasonally adjusted time series published
 - Little information on methodology
 - Often only rates of change available
- Few authors published research on seasonal and holiday effects for Chinese data

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Brief history of Chinese holidays

	Provisions on holidays
1949 to 1999	First version - New Year, Chinese New Year, Labour Day - National Day - 7 public holidays
From 2000	Introduction of Golden Weeks - Golden Weeks: Chinese New Year, Labour Day - National Day - 10 public holidays
From 2008	Rearrangements - Abolishment of Labour Day Golden Week - Introduction of traditional holidays - 11 public holidays

国务院办公厅关于2020年 部分节假日安排的通知

国办发明电〔2019〕16号

各省、自治区、直辖市人民政府，国务院各部委、各直属机构：

经国务院批准，现将2020年元旦、春节、清明节、劳动节、端午节、国庆节和中秋节放假调休日期的具体安排通知如下。

一、元旦：2020年1月1日放假，共1天。

二、春节：1月24日至30日放假调休，共7天。1月19日（星期日）、2月1日（星期六）上班。

三、清明节：4月4日至6日放假调休，共3天。

四、劳动节：5月1日至5日放假调休，共5天。4月26日（星期日）、5月9日（星期六）上班。

五、端午节：6月25日至27日放假调休，共3天。6月28日（星期日）上班。

六、国庆节、中秋节：10月1日至8日放假调休，共8天。9月27日（星期日）、10月10日（星期六）上班。

节假日期间，各地区、各部门要妥善安排好值班和安全、保卫等工作，遇有重大突发事件，要按规定及时报告并妥善处置，确保人民群众祥和平安度过节日假期。

国务院办公厅

2019年11月21日

Chinese holidays

Arrangement of holidays and workdays

General Office of the State Council on 2020

Notice of some holiday arrangements

State-owned invention electricity [2019] No. 16

[...]

1. New Year' s Day: Closed on January 1, a total of 1 day.
2. Chinese New Year: January 24 to 30 will be closed for 7 days.
Go to work on January 19 (Sunday) and February 1 (Saturday).
3. Qingming Festival: April 4-6 will be closed for 3 days.
4. Labor Day: May 1st to 5th will be closed for 5 days.
Go to work on April 26 (Sunday) and May 9 (Saturday).
5. Dragon Boat Festival: June 25 to 27 will be closed for 3 days.
Go to work on Sunday, June 28.
6. National Day and Mid-Autumn Festival: October 1-8 will be closed for 8 days.
Go to work on September 27th (Sunday) and October 10th (Saturday).

[...]

Office of the State Council
November 21, 2019

Chinese holidays

Arrangement of holidays and workdays

Mon	Tue	Wed	Thu	Fri	Sat	Sun
21 Sep	22 Sep	23 Sep	24 Sep	25 Sep	26 Sep	27 Sep
28 Sep	29 Sep	30 Sep	01 Oct	02 Oct	03 Oct	04 Oct
05 Oct	06 Oct	07 Oct	08 Oct	09 Oct	10 Oct	11 Oct
12 Oct	13 Oct	14 Oct	15 Oct	16 Oct	17 Oct	18 Oct

	Mid-Autumn Festival (1 October)
	National day (1-3 October)
	Additional holiday
	Additional workday

Chinese holidays

Overview

	Timing	Number of national holidays	
		2000-2007	From 2008
New Year ¹	1 January	1	1
Chinese New Year	January/February	3	3
Tomb-Sweeping-Day ¹	3,4 or 5 April		1
Labour Day ¹	1 May	3	1
Dragon Boat Festival ¹	May/June		1
Mid-Autumn Festival	September/October		1
National Day ¹	1 October	3	3
		10	11

¹Substitute workdays can be in different months.

Chinese holidays

Overview

	Timing	Consecutive work-free days	
		2000-2007	From 2008
New Year ¹	1 January	[1-3]	[1-3]
Chinese New Year	January/February	[7]	[7]
Tomb-Sweeping-Day ¹	3,4 or 5 April		[3]
Labour Day ¹	1 May	[7]	[3-5]
Dragon Boat Festival ¹	May/June		[3]
Mid-Autumn Festival	September/October		[3,8 ²]
National Day ¹	1 October	[7]	[3,8 ²]
		10	11

¹Substitute workdays can be in different months. ²If Mid-Autumn Festival and National Day overlap.

Chinese New Year



- Also known as: Spring Festival
- Visiting family ~→ Travel
- Traditionally ends with first full moon (15 days later)

Additional holidays



New Year

- Number of days off: 1-3
- Substitute work days sometimes in previous year



Labour Day

- Travel and leisure



National Day

- Family reunions
- Political and military ceremonial events

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Construction of regressors

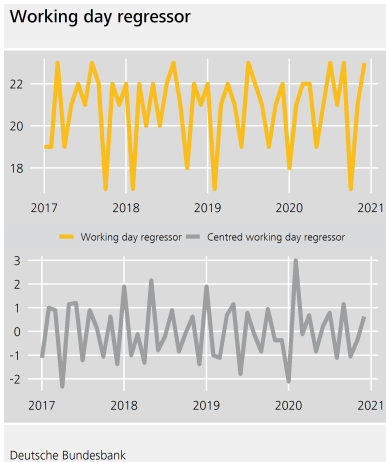
Reminder: RegARIMA

Results

Summary

Construction of regressors

Working day regressor

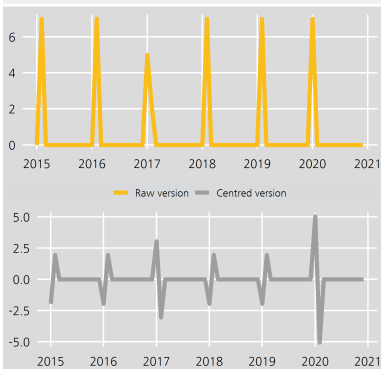


- For period t :
$$X_t = L_t - W_t - F_t + E_t$$
 - X_t : Number of working days
 - L_t : Total days
 - W_t : Number of weekend days (Saturday, Sunday)
 - F_t : Days in festival period
 - E_t : Substitute working days
- Centring on period-specific means

Construction of regressors

Chinese New Year

Regressor for Chinese New Year

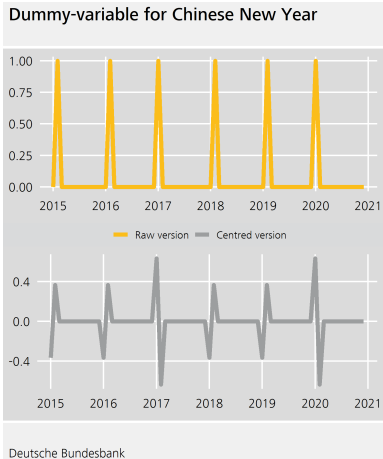


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- For period t :
 $X_t = \text{Number of days in period}$
- Centring on period-specific means

Construction of regressors

Chinese New Year - dummy variable



- For period t :

$$D_t = \mathbb{1}_{\text{Holiday in } t}$$

- Centring on period-specific means

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■ RegARIMA (I/V)

RegARIMA:

- Reg: Regression
- AR: Autoregressive
- I: Integrated
- MA: Moving Average

RegARIMA (II/V)

The parts of RegARIMA:

An AR(1) process:

$$y_t = \phi_1 y_{t-1} + \varepsilon_t$$

RegARIMA (II/V)

The parts of RegARIMA:

An AR(1) process:

$$y_t = \phi_1 y_{t-1} + \varepsilon_t$$
$$(1 - \phi_1 B)y_t = \varepsilon_t$$

RegARIMA (II/V)

The parts of RegARIMA:

$$\underbrace{(1 - \phi_1 B - \dots - \phi_p B^p)}_{\text{regular}} \underbrace{(1 - \Phi_1 B^{12} - \dots - \Phi_P B^{12*P})}_{\text{seasonal}} y_t = \varepsilon_t$$

$$\phi_p(B) \Phi_P(B^{12}) y_t = \varepsilon_t$$

RegARIMA (II/V)

The parts of RegARIMA:

An MA(1) process:

$$y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

RegARIMA (II/V)

The parts of RegARIMA:

An MA(1) process:

$$y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

$$y_t = (1 + \theta_1 B) \varepsilon_t$$

RegARIMA (III/V)

The parts of RegARIMA:

$$y_t = \underbrace{(1 + \theta_1 B + \dots + \theta_q B^q)}_{\text{regular}} \underbrace{(1 + \Theta_1 B^{12} + \dots + \Theta_Q B^{12*Q})}_{\text{seasonal}} \varepsilon_t$$

$$y_t = \theta_q(B) \Theta_Q(B^{12}) \varepsilon_t$$

RegARIMA (IV/V)

The parts of RegARIMA:

$$\underbrace{(1 - B)^d}_{\text{regular}} \underbrace{(1 - B^{12})^D}_{\text{seasonal}} y_t = \varepsilon_t$$

$$\nabla^d \nabla_{12}^D y_t = \varepsilon_t$$

RegARIMA (V/V)

The parts of RegARIMA:

$$y_t = \sum_{i=1}^r \beta_i x_{it} + \varepsilon_t$$

Complete RegARIMA model

Complete model

A seasonal **RegARIMA**($p \ d \ q$)($P \ D \ Q$) is given by

$$\phi_p(B)\Phi_P(B^{12})\nabla^d\nabla_{12}^D\left(y_t - \sum_{i=1}^r\beta_ix_{it}\right) = \theta_q(B)\Theta_Q(B^{12})\varepsilon_t$$

Prominent ARIMA models

Ollech and Webel (2022)

Rank	Non-seasonal		Seasonal	
	Model	Weight	Model	Weight
1	(011)	34.4	(011)(011)	47.5
2	(110)	13.4	(010)(011)	8.3
3	(000)	9.2	(311)(011)	7.7
4	(021)	8.4	(210)(011)	6.6
5	(101)	7.3	(110)(011)	6.4
6	(111)	5.6	(211)(011)	4.4
7	(100)	5.4	(012)(011)	4.0
8	(010)	4.9	(111)(011)	3.6
9	(210)	3.6	(011)(111)	2.5
10	(012)	3.2	(010)(100)	2.0

Finding the optimal ARIMA model (I/II)

Automatic choice

- Gomez-Maravall algorithm
- Hyndman-Khandakar algorithm

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Comparing models

- Compare to Airline model
- Use information criteria (AICc, BIC)
- Evaluate forecast errors
- Inspect forecasts visually

Finding the optimal ARIMA model (II/II)

Robustness checks

- Change span of time series
- Compare to aggregate or component series

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Robustness checks

- Change span of time series
- Compare to aggregate or component series

Diagnostics

- Look at ACF and PACF
- Look at significance of estimated coefficients
- Try simplification of model

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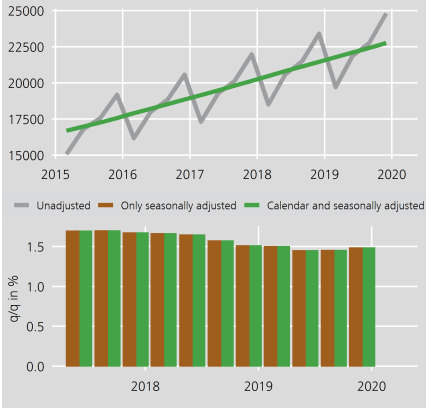
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GDP (volume)

GDP (volume)

Billion CN¥ of 2015 prices



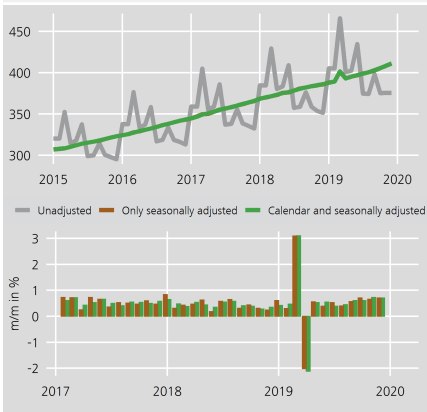
Source: NBS, Haver, own calculations
Deutsche Bundesbank

No calendar effects identified

Industry production

Industrial production

2015 = 100



Source: NBS, Haver, own calculations
Deutsche Bundesbank

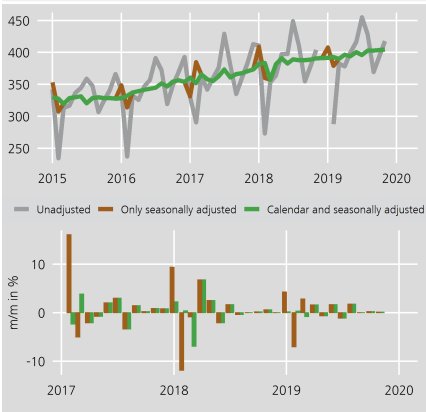
Semi-elasticity in % (SE)

Working days	0.08* (0.03)
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Electricity consumption by industry

Electricity consumption by industry

Billion KW/H



Source: CEC, Haver, own calculations
Deutsche Bundesbank

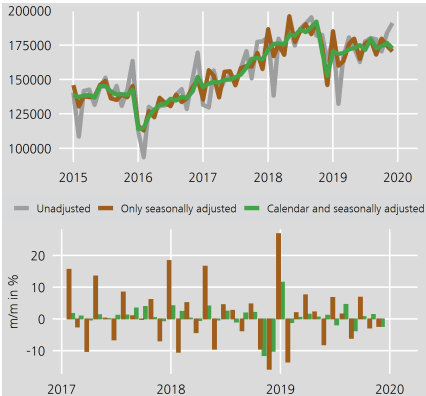
Semi-elasticity in % (SE)

CNY	-2.06*** (0.21)
CNY, 1w before	-0.67*** (0.16)

Merchandise imports

Merchandise imports

Million US\$, cif



Source: CC, Haver, own calculations
Deutsche Bundesbank

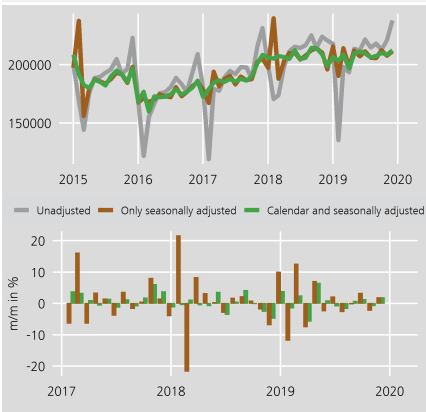
Semi-elasticity in % (SE)

Working days	3.27*** (0.25)
CNY	-0.98*** (0.24)

Merchandise exports

Merchandise exports

Million US\$, fob



Source: CC, Haver, own calculations
Deutsche Bundesbank

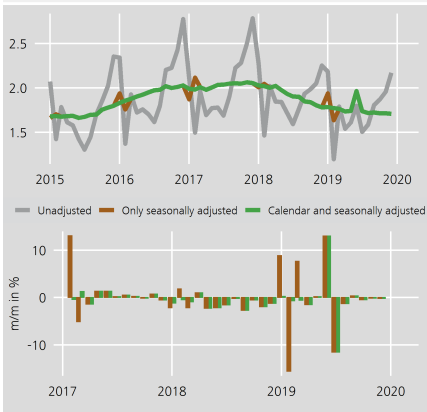
Semi-elasticity in % (SE)

Working days	1.09** (0.29)
CNY, 2w before	0.78*** (0.13)
CNY, 2w after	-1.95*** (0.20)

Sales of passenger cars

Sales of passenger cars

Million units



Source: CPCA, Haver, own calculations
Deutsche Bundesbank

Semi-elasticity in % (SE)

CNY	-3.39*** (0.27)
CNY, 2w before	0.84*** (0.08)

Identification of seasonal effects

Overview

Indicator	Span	P-value	
		QS-test	Kruskall-Wallis test
Merchandise imports	2002-2009	<0.001	<0.001
	2010-	<0.001	<0.001
Merchandise exports	2002-2009	<0.001	<0.001
	2010-	0.000	<0.001
Industry production	2002-2007	0.000	<0.001
	2010-	0.000	0.000
Electricity consumption by industry	2010-	0.000	<0.001
Sales of passenger cars	2014-	0.000	<0.001
GDP (volume)	2002-	0.000	<0.001

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The summary

- Working days and Chinese New Year are most important calendar influences
- Seasonal effects in all time series considered here
- Calendar and seasonal adjustment supports economic analysis